

Electromagnetic Force	Responsible for interaction between charged particles	Virtual Photon	Infinite Range
Gravitational Force	Responsible for interaction between massive particles	Graviton (theoretical)	Infinite Range
Weak Nuclear Force	Responsible for the decay of particles	W ⁺ , W ⁻ , Z ⁰ Boson	Subatomic Range
Strong Nuclear Force	Responsible for keeping the atomic nucleus together, acts on Hadrons only	Gluons, Pions	Repulsive < 0.5 fm 0.5 fm < attractive < 3.0 fm 3.0 fm < Negligent

Anihilation

When a particle and its antiparticle collide, they annihilate to produce two photons that have a total energy of $E = mc^2 + \frac{1}{2}mv^2$ and move in opposite direction

Pair Production

A gamma ray of minimum energy of $E = 2mc^2$ may produce a particle antiparticle pair that will move in opposite direction to each other

Antimatter: A type of particles with opposite charge, baryon number, and strangeness to their matter counterparts (but same rest mass). They annihilate upon contact with each other and are produced in pairs.

Same, same, but different

Alpha Decay	${}^A_ZX \rightarrow {}^{A-4}_{Z-2}Y + {}^4_2\alpha$
Beta-Minus Decay	${}^A_ZX \rightarrow {}^A_{Z+1}Y + {}^0_{-1}\beta + \bar{\nu}_e$
Beta-Plus Decay	${}^A_ZX \rightarrow {}^A_{Z-1}Y + {}^0_{+1}\beta + \nu_e$
Gamma Decay	${}^A_ZX \rightarrow {}^{A-4}_{Z-2}Y + {}^4_2\alpha + \gamma$
Electron Capture	$P + e \rightarrow n + \nu_e$
Electron-Proton Collision	

Photoelectric Effect: When light of frequency greater than the threshold frequency is incident on the surface of a metal, electrons are liberated from the surface of the metal

Threshold Frequency: Photoelectric effect only occurs if incident light has a frequency greater than this

Work Function: The minimum energy required to emit an electron from a metal surface

Maximum Kinetic Energy: This is the difference between energy of the incident photons and the work function

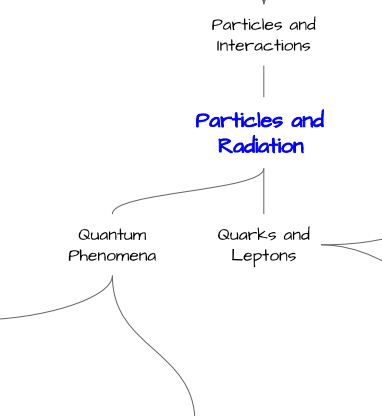
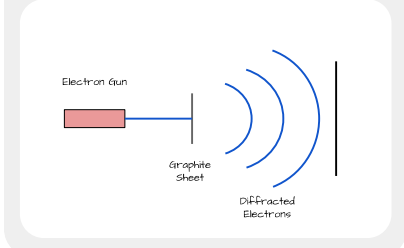
Stopping Potential: This is the pd that needs to be induced to match E_{max} of the photoelectrons and halt them from being emitted. This is found by $E = eV_s$

The Photoelectric Effect proves that particle nature of light as the PE effect only occurs at certain range of frequencies if it was only caused by energy supplied increasing the intensity of incident EMR even under the threshold frequency, PE must be liberated. However, as the energy supplied is linked to the frequency, and not intensity, a 1-1 relationship between light and electrons is shown, proving light is a particle

It has been known for long that light has a dual-nature between waves and particles. According to De Broglie, electrons do too.

De Broglie Wavelength: This is the equation for calculating the wavelength of a moving particle to achieve wave-nature.

Electron Diffraction: Electrons travelling in their De Broglie Wavelength, can diffract when passed through atomic gaps of a crystal. This is proof of wave-particle duality in electrons



Ionisation: This is the process of which an atom gains or loses an electron resulting in a net charge.

Excitation: This is the process when an electron gains energy and 'jumps' to a higher energy level.

Excitation by collision

Ionisation by collision

Excitation by photon absorption

De-excitation by photon emission

Absorption Spectra: This is a spectrum with gaps of frequencies where light has been absorbed.

Emission Spectra: This is a spectrum of only lines from parts of the spectrum where light was emitted

Electron Volt: Work done to accelerate an electron through one Volt. $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

Specific Charge: The charge of an atom per unit mass (C/kg)

Mass Number (A): No. of Nucleons

Proton Number (Z): No. of Protons

Isotopes: A variant of an element with a different mass number (neutrons)

Fundamental Particles: A particle that cannot be further broken down to constituent parts: Quarks, and Leptons

Strangeness: Strange particles take longer to decay and hold strange quarks. Created in pairs, decays only in decrements of one

Conservation Laws: These are attributes that are conserved during an interaction:

- Charge is always conserved
- Baryon number is always conserved
- Lepton number (of the same type) is conserved
- Strangeness is conserved, except during a decay

Particles

Hadron

Non-fundamental particles that are composed of quarks (Particles that experience the Strong Force)

Baryons

Hadrons with three quarks in form qqq . The antiparticle has a form $\bar{q}\bar{q}\bar{q}$

Protons
Neutrons

Mesons

Hadrons with two quarks in form $q\bar{q}$. The antiparticle has the same yet opposite form

Pions (π^+ , π^- , π^0)
Kaons (K^+ , K^- , K^0)

Quarks

Fundamental Particles that make-up Hadrons.

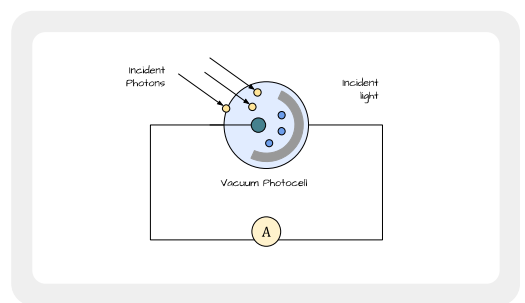
Up	Down	Strange
Top	Bottom	Charm

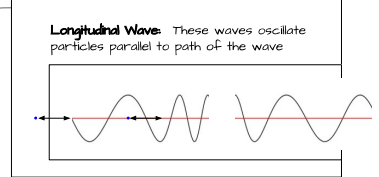
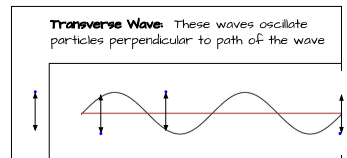
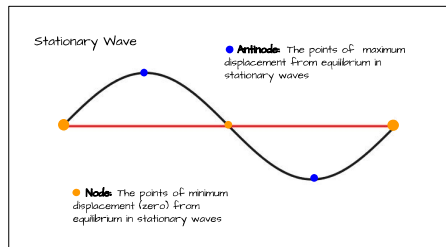
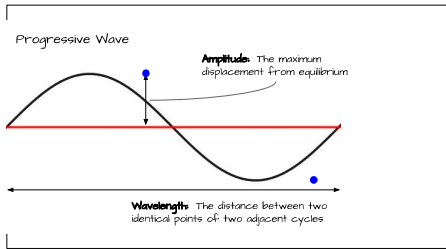
Leptons

Fundamental particles with very little mass. Can decay into other lepton types.

Electron
Muon
Tauon
Electron Neutrino
Muon Neutrino
Tauon Neutrino

p	uud
n	udd
π^+	$u\bar{d}$
π^-	$\bar{u}d$
π^0	$\frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d})$
K^+	$u\bar{s}$
K^0	$\frac{1}{\sqrt{2}}(u\bar{s} + d\bar{s})$
K^-	$\bar{u}s$



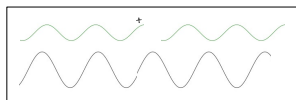


In-Phase: This is when the phase difference is 0 rad, or 0 degrees.

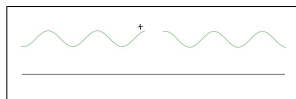
Out of Phase: This is when the phase difference is not 0 rad, or 0 degrees.

Anti-Phase: This is when the phase difference is π rad, or 180 degrees.

Constructive Interference: This is the process of two waves with the same vector displacement combine.



Destructive Interference: This is the process of two waves with displacement in the opposite directions. If the displacements have an equal absolute value, total destructive interference occurs.



Phase difference: This is the difference between two waves based on their stage in the wave cycle. Assume a cycle is a circle of radius one, hence circumference is 2π . This is the concept of radians.

Superposition: This is the process of two waves combining as they pass each other, causing a resultant displacement with the vector sum of the constituent waves. This causes **Interference**.

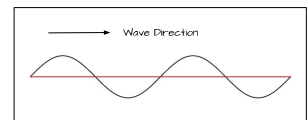
Polarisation: This is when a wave only oscillates in one plane. This only occurs with transverse waves.

Coherent: This is a source of waves where the wavelength, frequency are the same and phase difference is constant.

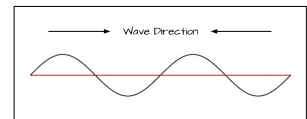
Types by direction of oscillation

Types by energy transfer

Progressive Wave: These waves transfer energy without transferring matter and is caused by oscillating particles.



Stationary Wave: These waves are formed by the superposition of two coherent and opposite progressive waves.



First Harmonic: The lowest frequency where a stationary wave forms. The distance between the nodes is $\frac{1}{2}\lambda$.

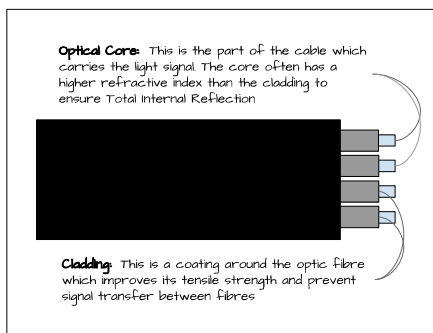
$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

Wave Speed: The speed of which a wave transfers energy. $c = f\lambda$.

Frequency: The rate of completion of wave cycles. $f = 1/T$.

Period: The time taken by a wave to complete one full wave-cycle.

Waves and Optics



Pulse Broadening: This is the process of the signal elongating when passed down the optic cable due to dispersion.

Modal Dispersion: This is caused by waves entering the cable at slightly different angles, causing them to reach the end at slightly different times.

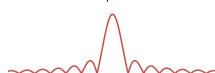
Material Dispersion: This is caused by waves of different λ travelling at slightly different speeds causing them to reach the end at slightly different times.

Optical Fibre

Properties of light

Optics

Single Slit Interference Intensity Pattern



Young's Double Slit Intensity Pattern



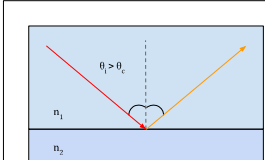
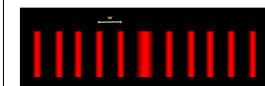
Diffraction Grating Intensity Pattern



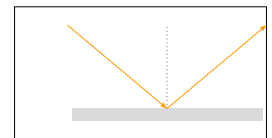
Diffraction: This is the property of waves dispersing when they pass through a slit of similar size to their wavelength.

Diffraction Grating: A sheet with hundreds of slits per millimeter which results in sharper interference patterns due to $d \sin \theta = n\lambda$.

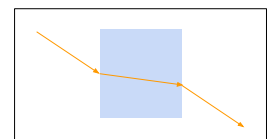
Fringe Spacing: This is the distance between two adjacent maxima or minima. Calculated using $w = \lambda D / s$.



Reflection: This is the property of light being redirected in the same angle as it was incident to the normal.



Refraction: This is the property of light being changing direction when entering a new material to adjust to its new speed.



Refractive index: This is the ratio between speed of light in a vacuum and the speed of light within a material. $n = c/v$.

Snell's Law: This shows a relationship between refractive indices and angles of incidence and refraction. $n_1 \sin \theta_1 = n_2 \sin \theta_2$.

Critical Angle: This is the minimum angle of incidence where (for rays travelling from a denser medium to a lighter medium) the ray is not refracted, but reflected inwards. $\sin \theta_c = n_2 / n_1$ (for $n_1 > n_2$).

Total Internal Reflection: This is the process of rays reflecting at the boundary where $n_1 > n_2$ and the angle of incidence is greater than the critical angle.

Tilting: The process of exerting a turning force on an object about a point.

Topping: This is when the line of action of weight (which runs through the centre of mass) passes the pivot point.

Equilibrium: This is a state of an object where the resultant forces equal to zero

Moment: This is the product of the force and the perpendicular distance from the line of action of the force to the point. (moment = $F \times d$)

Principle of Moments: For equilibrium the sum of clockwise moments equals the sum of anticlockwise moments: $\Sigma \text{moments (CW)} = \Sigma \text{moments (ACW)}$

Centre of mass: This is point of an object where all mass is thought to be concentrated / point where no single turning force can cause a turning effect

Couple: A pair of equal and opposite forces that are acting on along two distinct lines of action. These exert a turning force on the object with a total moment that is the same at any given point.

Vectors using Scale Diagrams: Draw a scale diagram conserving direction and magnitude with each arrow end linked to another arrow's head.

Perpendicular vectors using calculation:

The magnitude of the resultant is derived using: $(F_1^2 + F_2^2)^{1/2} = F_R$

The angle $\theta = \tan^{-1} (F_1 / F_2)$

Resolving into perpendicular vectors: Draw F as a diagonal of a parallelogram and then complete it. The height is F_V and the width is F_H

This can be solved mathematically too: For the first direction from the anticlockwise, it is $\sin \theta$ and for the other it is $\cos \theta$ (where θ is angle from positive x-axis)

Displacement: The linear distance in a given direction

Speed: The rate of change of distance: $v = \Delta d / t$

Velocity: The rate of change of displacement: $v = \Delta s / t$

Acceleration: The rate of change of velocity: $a = \Delta v / t = (v - u) / t$

Free Fall: This is when an object is released and accelerates downwards with no force except its own weight. Acceleration due to free fall on Earth is 9.81 ms^{-2}

Projectile Motion: This is the motion of an object where gravity is the only force acting on it. The vertical acceleration is g and there is no acceleration in the horizontal. The vertical and horizontal components are independent to each other.

Momentum: This is the product of the mass and velocity of an object. momentum = $m v$

Conservation of Momentum: This principle states that for a system of interacting objects, the total momentum is constant given no external forces act on the system. $\Sigma \text{mom}_i = \Sigma \text{mom}_f$

Impulse: The change of momentum of an object across time. $Ft = \Delta(mv)$

Explosion: The process where objects at rest fly apart due to some process. The vector sum of all momentum would still be equal to zero $\Sigma \text{momentum} = 0$

Collision: This is the process of two objects colliding against each other.

Elastic Collision: This is when there is no loss of kinetic energy during the impact and rebound. $E_{k(i)} = E_{k(f)}$

Inelastic Collision: This is when there is some loss of kinetic energy during the impact and rebound. $E_{k(i)} > E_{k(f)}$

Perfectly Inelastic Collision: This is when all the kinetic energy is lost during the impact and there is no rebound. $E_{k(f)} = 0$

On the Move

$$v = u + at$$

$$s = \frac{1}{2} (u + v) t$$

$$s = ut + \frac{1}{2} at^2$$

$$v^2 = u^2 + 2as$$

Forces in Equilibrium

Vector: A physical quantity with magnitude and a direction. Can be represented in a column vector or i and j notation.

Scalar: A physical quantity with only a magnitude and no direction.

Newton's Laws of Motion

Mechanics and Materials

Force and Momentum

Conservation of Energy: This is a principle that states that energy is never created nor destroyed, only transferred.

Work: The product of force and distance travelled in direction of energy.

Kinetic Energy: The energy of an object gained via motion. $E_k = \frac{1}{2} mv^2$

Potential Energy: The energy of an object gained via its position. $E_p = mg\Delta h$

Power: The rate of transfer of energy. $P = \Delta E / \Delta t = \Delta W / \Delta t$

Efficiency: The ratio of useful energy output to total energy input.

Work, Energy and Power

Materials

First Law: If the net force on an object is zero, it will have a constant velocity.

Second Law: The total force acting on an object is proportional to the rate of change of momentum of the said object. $F = ma = \Delta mv / t = m(v - u) / t$

Third Law: For every force exerted on a body (A) by another (B), A will exert a force equal in magnitude, type but opposite direction on B.

Weight: The force exerted by an object due to gravitational acceleration.

Inertia: This is the resistance of an object to a change in its motion.

Terminal Velocity: This is the maximum velocity an object can achieve, moving through a fluid due to its acceleration and fluid resistance being equal to each other.

Density: A physical quantity which is mass per unit volume of an object ($\rho = m/v$)

Hooke's Law: The force needed to stretch a spring is directly proportional to the extension of the spring from its natural length. ($F = k\Delta L$)

Spring Combinations

Springs in Parallel: $k = k_1 + k_2 + \dots$

Springs in Series: $1/k = 1/k_1 + 1/k_2 + \dots$

Elastic Potential Energy: $E_p = \frac{1}{2} F\Delta L = \frac{1}{2} k\Delta L^2$

Elasticity: The property of an object to be deformed and regain its original shape. Deformation that stretches is tensile and deformation that compresses is compressive.

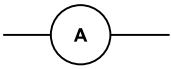
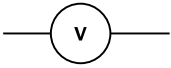
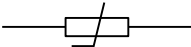
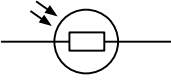
Tensile Stress: The tension in the wire per unit area. $\sigma = T/A$

Tensile Strain: The ratio of the extension of the wire to the original length of the wire. $\epsilon = \Delta L/L$

Young Modulus: This is a constant of all materials which is calculated using $E = \sigma/\epsilon$

Elastic Limit: This is the maximum extension of an object before suffering plastic deformation.

Ultimate Tensile Stress (UTL)/ Breaking Stress: This is the maximum stress that can be applied on an object before it loses its strength, extends, and becomes narrower at its weakest point, eventually leading to breakage.

Ammeter	This is a device which measures the current of a circuit. This is connected in series and ideally should have zero resistance	
Voltmeter	This is a device which measures the pd of a circuit. This is connected in parallel and ideally should have infinite resistance	
Thermistor	This is a temperature sensitive resistor where the resistance reduces when temperature increases.	
Light Dependent Resistor	This is a light sensitive resistor where the resistance reduces when light intensity increases.	

Potential Difference: This is the work done per unit charge. $V = W/Q$

Electromotive Force (EMF): This is defined as the electrical energy produced per unit charge passing through the source. $\mathcal{E} = I(R + r) = E/Q$

Current: The rate of flow of charge in a circuit. $I = \Delta Q / \Delta t$

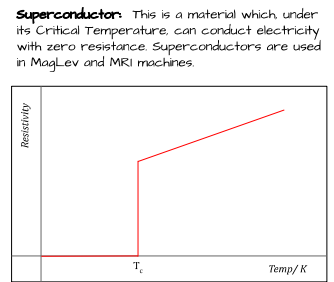
Power: The rate of transfer or conversion of energy. $P = IV = I^2 R = V^2 / R$

Resistance: The ratio between the potential difference and the charge (i.e. How difficult it is for current to flow). $R = V/I$

Resistivity: A material-specific constant which is the resistance across a cross-sectional area of a wire per unit length. $\rho = RA/L$

Ohm's Law: A principle that states that the pd across a conductor is directly proportional to its current given it's held in constant physical conditions. The constant being resistance.

Ohmic Conductor: This is a conductor which obeys Ohm's law



Semiconductor: This is a material where its conductivity increases with temperature.

	Series	Parallel
Potential Difference	$V_{\Sigma} = V_1 + V_2 + V_3 + \dots$	$V_{\Sigma} = V_1 = V_2 = V_3 = \dots$
Current	$I_{\Sigma} = I_1 = I_2 = I_3 = \dots$	$I_{\Sigma} = I_1 + I_2 + I_3 + \dots$
Resistance	$R_{\Sigma} = R_1 + R_2 + R_3 + \dots$	$1/R_{\Sigma} = 1/R_1 + 1/R_2 + 1/R_3 + \dots$

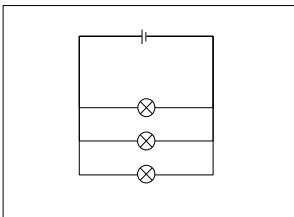
Potential Divider: This is a method of splitting the potential difference by connecting two resistors in series. The total potential difference is split in the ratio of their resistances

Electric Current

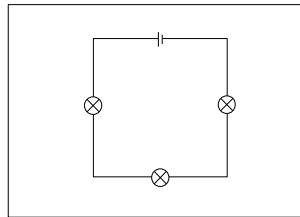
Electricity

DC Circuits

Parallel Circuits: Components are connected in parallel loops which gets them in separate loops.

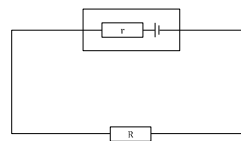


Series Circuits: Components are connected in series, within one loop.



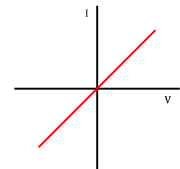
Terminal Potential Difference: This is the pd across the power source. $\mathcal{E} = V = Ir$

Internal Resistance: This is the resistance of the power source.

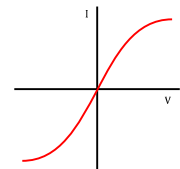


I-V Diagrams

Ohmic Conductor



Filament Lamp



Diode

